

SGA 5: ℓ -adic Cohomology and L -Functions

Draft English translation of the volume introduction

Introduction

This volume brings together the exposés from Grothendieck’s seminar at the IHES in 1965–66, initially distributed in the form of mimeographed notes. Compared with the original version, the only important changes concern Exposé II, which is not reproduced, and Exposé III, which has been entirely rewritten and enlarged by an appendix numbered III B. Apart from a few detailed modifications and added footnotes, the other exposés have been left as they stood.

The heart of the seminar is the Lefschetz formula in étale cohomology (III, III B, XII), and its application to the cohomological interpretation of L -functions (XIV). All the results announced by Grothendieck in his Bourbaki seminar talk [2] are proved here completely. The trace formulae established in (III, III B, and XII), by different methods, are more general than is necessary for proving merely the rationality of the L -functions. A complete and much shorter proof of the latter appears in Deligne’s exposé (SGA 4^{1/2}, Report), where Grothendieck’s method from Exposés XII and XIV is followed, but stripped of all superfluous generality. We nevertheless hope that the formulae of III and III B may be useful in other situations. As for the rest of the seminar, it consists of two exposés on the theory of passage to the limit that gives rise to ℓ -adic cohomology (V, VI), and of various complements to the duality formalism (I, VII) and to the Lefschetz formula (VII, X).

Here, more precisely, is the content of this volume.

Exposé I is independent of the rest of the seminar. Its main result is that, on a regular scheme satisfying certain additional local hypotheses, and assuming the availability of resolution of singularities and the purity theorem, the constant sheaf with value $\mathbb{Z}/n\mathbb{Z}$, for n prime to the residue characteristics, is dualizing (I.3.4.1). This theorem applies in particular to excellent regular schemes of characteristic zero, thanks to Hironaka and to Artin’s results (SGA 4, XIX). The case of a regular scheme of dimension 1 is treated separately, by another, very elementary method; see also SGA 4^{1/2}, Duality. Let us also point out that, by a different argument using neither resolution nor purity, Deligne proves in SGA 4^{1/2}, Finiteness Theorem, that if $f : X \rightarrow S$ is a scheme of finite type over a regular scheme of dimension 0 or 1, then the complex $f^!(\mathbb{Z}/n\mathbb{Z})_S$, for n prime to the residue characteristics, is dualizing. It is not known, however, whether this result, which is more useful than the biduality theorem of Exposé I, and which overlaps it without containing it, extends to the case where the base scheme S is excellent, regular, and of dimension > 1 .

As was said above, Exposé II, entitled “Künneth formulae for cohomology with arbitrary supports”, does not appear in this volume. Written by L. Illusie from Grothendieck’s handwritten notes, it was devoted to theorems on cohomological properness and generic local acyclicity; but these were proved only under resolution hypotheses. A proof of the same results without these hypotheses, obtained since then by Deligne (SGA 4^{1/2}, Finiteness Theorem), made publication of that exposé unnecessary. Certain complements, concerning for example the case of normal-crossing divisors, were incorporated into an appendix to *loc. cit.* We add that theorems analogous to those of *loc. cit.*, in the case of sheaves of sets and noncommutative groups, are proved in Mme Raynaud’s exposé (SGA 1, XIII).

Exposé III contains a proof of the Lefschetz–Verdier formula for cohomological correspon-

dences between complexes of sheaves on schemes of finite type over a field. In Exposé III B, one shows how to compute the local terms of the formula in certain cases, for instance that of the Frobenius correspondence on curves. We refer the reader to the introductions of Exposés III and III B for more details on their contents.

Exposé IV, by A. Grothendieck, devoted to the construction of the cohomology class associated with a cycle, was written up by P. Deligne and published in SGA 4^{1/2}.

After technical preliminaries in Exposé V on adic inverse systems, Exposé VI defines the notion of a constructible $\mathbb{Z}_\ell$ -sheaf, respectively a constructible twisted constant sheaf, that is, a “smooth” sheaf in Deligne’s terminology ([1], SGA 4^{1/2}, Arcata). It then extends to constructible $\mathbb{Z}_\ell$ -sheaves, by passage to the limit from $\mathbb{Z}/\ell^n$ -sheaves, the formalism of higher direct images with proper supports. Exposé VI does not, however, give a definition of a derived category of $\mathbb{Z}_\ell$ -sheaves, with operations $\otimes^L$ and RHom extending those available in the derived category of $\mathbb{Z}/\ell^n$ -sheaves. A fortiori, the extension to $\mathbb{Z}_\ell$ -sheaves of the duality formalism of SGA 4, XVIII is not considered. This question was the subject of Jouanolou’s unpublished thesis. The reader may nevertheless consult the beginning of [1], where the sheaves $\underline{\mathrm{Ext}}^i$ of $\mathbb{Z}_\ell$ -sheaves are defined.

Exposé VII, independent of the rest of the seminar, is undoubtedly one of the most interesting. It contains: (a) the computation of the cohomology of several standard varieties, namely affine spaces, projective spaces, and flag varieties; (b) an account of the theory of Chern classes in étale cohomology; (c) a proof of the self-intersection formula

$$i^* i_* x = x c_d(N)$$

in étale cohomology, respectively in the Chow ring, the proof in the latter case being due to Mumford, together with various applications of this formula, including computation of the cohomology of blow-ups and the Gauss–Bonnet formula.

Exposé VIII contains sorites on finiteness conditions in derived categories and Grothendieck groups. These questions are taken up again in a more general framework in SGA 6, I and IV.

Exposé IX, by J.-P. Serre, entitled “Introduction to Brauer theory”, was published in [4] and is not reproduced in this volume. One of its principal results, due to Swan, concerning the existence of a projective module having as character the “Swan character”, is used in Exposé X, where an Euler–Poincaré formula is proved for a sheaf on a curve, slightly more general than the one appearing in Raynaud’s exposé [3].

Exposé XI, written by I. Bucur, does not exist: it was lost during a move, and the author had no copy of it. It was devoted to Grothendieck’s theory of commutative traces, generalizing that of Stallings [5]. A more sophisticated version of this theory, in a sheaf-theoretic framework, is presented in (III B 5, 6). Applications to Lefschetz formulae are given in Exposés XII and III B. The trace formula of Exposé XII is proved independently of the general formula of Exposé III, but it is shown in (III B 6) that the local terms appearing there are indeed those of the general formula, and that the latter implies it.

The absence of Exposé XIII is due only to numbering: the contents of Exposés X through XII had initially been planned to extend over four exposés.

Exposé XIV, sometimes also numbered XV, contains generalities on the Frobenius correspondence, including in particular the comparison between “geometric” Frobenius and “arithmetic” Frobenius. Starting from the trace formula of Exposé XII, or from III B, it gives Grothendieck’s proof of the rationality of L -functions. The two important points are reduction to the case

of curves, and the passage to the limit that makes it possible to reduce to a formula for $\mathbb{Z}/\ell^n$ -sheaves.

During the oral seminar, Grothendieck gave an exposé on open problems and stated a few conjectures. That exposé, which was intended to close the seminar, was unfortunately not written up, nor was his very beautiful introductory exposé, which surveyed the Euler–Poincaré and Lefschetz formulae in various contexts: topological, complex analytic, and algebraic.

I thank P. Deligne for having convinced me to write, in a new version of Exposé III, a proof of the Lefschetz–Verdier formula, thereby removing one of the obstacles to the publication of this seminar. I am very grateful to him for the improvements in exposition that he suggested to me, and for the help he gave me in preparing this volume for the publisher. I also thank Mme Lièvremont, who typed Exposés III and III B carefully and in a very short time.

I cannot end this introduction without paying tribute to the memory of I. Bucur, who died of cancer in September 1976. Those who, like me, had the privilege of having him as a friend will not forget his exquisite kindness, his smiling humor, and the charm of his conversation.

Paris, 19 February 1977
Luc Illusie

Continuation anchor. The next SGA 5 translation unit begins with Exposé I, “Dualizing complexes”, in the same source file, after the volume bibliography and list of SGA volumes.